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Quiz: Languages

■ $S = \{ \text{John, Jill, drives, jogs, carelessly, rapidly, frequently} \}$

■ $N = \{ \text{sentence, noun, verbphrase, verb, adverb} \}$

■ $V_0 = \text{sentence}$

■ Define $\rightarrow$

sentence $\rightarrow$ noun verbphrase

noun $\rightarrow$ John

noun $\rightarrow$ Jill

verbphrase $\rightarrow$ verb adverb

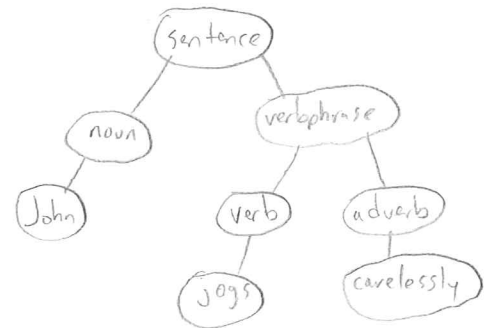
verb $\rightarrow$ drives

verb $\rightarrow$ jogs

adverb $\rightarrow$ carelessly

adverb $\rightarrow$ rapidly

adverb $\rightarrow$ frequently



1) Is "John jogs carelessly" a valid sentence, if so give the parse tree.

2) Is "Jill frequently jogs" a valid sentence, if so give the parse tree. no

3) List the language $L(G)$, i.e. all possible sentences (as a regular expression).

■ $G = (V, S, v_0, \rightarrow)$

■ $V = \{v_0, v_1, x, y, z\}, S = \{x, y, z\}$

■ $\rightarrow:$ $v_0 \rightarrow x v_0$

$v_0 \rightarrow y v_1$

$v_1 \rightarrow y v_1$

$v_1 \rightarrow z$

$x^* y y^* z$

4) What type of grammar is it?

Type 3